## 7.6 Highest Common Factor

We have learnt in class VI that iteration is a process wherein a set of instructions or structures are repeated in a sequence for a specified number of times or until a condition is met. Here, we are going to learn to find HCF by listing all factors and find the biggest, then to find HCF by repeated subtraction and see to how much faster the iteration goes (and how in fewer steps you get the HCF) and then how to improve further by repeated division and remainder and that both lead to the same solution but one is faster than the other.

We know that, HCF is used in simplifying or reducing fractions. To understand how this concept applies in real life, imagine the following situation.

### 7.6.1 Methods to find HCF (Highest Common Factor):

#### 1. Factorisation Method:

***Situation:***



Let us assume that you have 20 mangoes and 15 apples. You want to donate them together equally among the orphan children. How many orphan children can you help at the maximum?

Here, basically question demands finding HCF of two numbers. HCF is Highest Common Factor, also known as GCD (Greatest Common Divisor). HCF of two or more than two numbers is such that, it is the largest possible number which divides given numbers completely.

Here, let us find the HCF of 20 mangoes and 15 apples.

Factors of 20 = 1,2,4,(5),10,20

Factors of 15 = 1,3,(5),15

So, the HCF of 20 and 15 is 5. That is, you can a help maximum of 5 orphan children.

So that for **5 children** you can give **4 mangoes** (20 ÷ 5 = 4)  and **3 apples** (15 ÷ 5 = 3)  to each of them. In this way you can distribute equally the mangoes and the apples to each child.

#### 2. Prime Factorisation Method:

***Situation:***

Suppose there are 18 students in Class VII and 27 in Class VIII and each class is divided into teams to prepare for an upcoming sports tournament, with the winning teams from each class play each other in the final. What would be the biggest possible team size that you could divide both these classes such that each team has exactly the same number of students and that no one is left behind.



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